David DUNEAU, PhD

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University of Lisbon (FCUL)
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French (native); English (fluent)
R (fluent)

Research interests

I recently joined the University of Lisbon (cE3c) as a PI. As an evolutionary infection biologist, I integrate experimental and theoretical approaches to understand how host heterogeneity shapes pathogen evolution. I demonstrated that understanding infection requires examining sequential steps rather than single outcomes. Applying novel statistical methods to within-host dynamics, I revealed that early infection events determine survival and that pathogens face trade-offs between stages. My current research program investigates how host heterogeneity, particularly sexual dimorphism, drives pathogen plasticity and evolution, testing whether parasites actively respond to and adapt to variation in host environments. Working across Drosophila, birds, humans, and bacterial systems, I combine GWAS, experimental evolution, transcriptomics, and mathematical modeling to reveal fundamental principles of host-pathogen interactions.

Vertebrate & invertebrate immunity ◘ Genetic basis of quantitative traits ◘
◘ Sexual dimorphism ◘ Within-host pathogen dynamics ◘
◘ Biostatistics ◘ Bioinformatics ◘

Research positions

2025 – 2031:
FCT Research fellow (CEEC)
Centre for Ecology, Evolution, & Environmental Changes (cE3c), University of Lisbon, PT.

2023 - 2025:
Senior post-doctoral research associate
Bénézech Lab, Centre for Cardiovascular Science, University of Edinburgh, UK.

2021 - 2023:
Visiting researcher at the Institute of Ecology and Evolution, University of Edinburgh, UK.

2020 - 2021:
Independent researcher, *Laboratoire International Associé* programme
Instituto Gulbenkian de Ciência, Portugal & University Toulouse 3, France.

Apr. 2015 to Dec. 2019:
Senior post-doctoral research associate
Lab. Evolution and Biological Diversity (EDB) University Toulouse 3.

Nov. 2011 – Dec. 2014:
SNSF post-doctoral research associate
Lazzaro Lab, Cornell University, USA.

Career break

01/11/2017 to 31/08/2019: ~2 years (22 months) due to parental responsibilities (1 child now 10 years old).

Education

2007 – 23/09/2011: **PhD student in Evolutionary parasitology** Basel Univ. (CH) (Sept 23, 2011)
'Evolutionary and proximate mechanisms shaping host-parasite interactions: The case of Daphnia magna *and its natural bacterial parasite* Pasteuria ramosa.*'*

2004 - 2006: **Masters in Ecology and Evolutionary Biology**, Montpellier Univ. (FR)

2001 - 2004: **Bachelor of Science in Organismal Biology**, Montpellier Univ. (FR)

Funding

- Principal Investigator fellowships (6 years PI position)
    - PI fellowship Concurso Estímulo ao Emprego Científico (CEEC) Individual (2025-2031)
- Postdoc fellowships (278K€)
    - Post-doctoral fellowship from Gulbenkian Foundation in Lisbon (2020 & 2021; 30K€)
    - Post-doctoral '*Prestigious and Marie Curie Fellowship*' (2017; 51K€)
    - Post-doctoral fellowship from ANR, French LabEx TULIP (2015; 112K€)
    - Post-doctoral fellowship from Swiss NSF (2011 & 2013; 76K€)
    - PhD extension fellowship from the Emilia Guggenheim-Schnurr foundation in Basel (2009; 9K€)
- Grants
    - TENOVUS Scotland; 2025-2026; (£23,5K) - **co-PI** with Haythorne L. - *Investigation of how chemicals from plastic pollution increase the incidence of type 2 diabetes*
    - FCT Portugal; 2025-2028; (£250K) - **co-PI.** with Beldade P. - *Thermal development plasticity in adult immunity: molecular mechanisms and evolutionary consequences.*
    - BHF REA4 Pump-Priming; 2025 (£35K) – **co-PI.** with Haythorne L. - *How do endocrine disrupting chemicals, from plastic pollution, impact glucose metabolism of insulin secreting cells?*
    - CSE-CMVM Immune Regulation Pump-Priming; 2025 (£9K) **co-I.** with Regan J., Bénézech C., Jenkins S. - *Sex as a regulator of macrophage function during infection.*
    - *PacBio* Win Free sequencing Competition; 2024 (£5K) **PI.** - Constructing metagenomic databases for oral microbiomes of non-model species.
    - EDB Toulouse Univ. funded by ANR, French LabEx CEBA; 2016 (20K€) **co-PI.** with Zinger L. - *Disentangling the factors shaping gut microbiota diversity across arthropod predators.*
    - EDB Toulouse Univ., funded by "New frontiers" LabEx TULIP project; 2016 (82K€) **co-PI.** with Ferdy J-B - *Pathogens adaptation to their host's microbiome*.
    - FCT Portugal; 2016; (192K€) **co-PI.** with Beldade P. *Adaptive Developmental Plasticity: genetic and environmental components of phenotypic variation*.
- 5 Travel grants to attend conferences (from hosting universities, ~500$ each).

Main scientific accomplishments

I have built an international reputation in evolutionary biology of disease, working at leading institutions (Cornell, Edinburgh, Toulouse, Gulbenkian) and publishing 37 papers, with 6 preprints and 4 manuscripts in preparation. Using primarily Drosophila as a model system, I integrate experimental and computational approaches to understand host-pathogen interactions.

Major research contributions:

Sexual dimorphism in infections

- Proposed and demonstrated that parasites can adapt to the sex of host they encounter most frequently, behaving differently depending on host sex (Duneau & Ebert PLoS Biol 2012; Duneau et al. BMC Biol 2012; Rubinic et al. in prep)
- Uncovered mechanisms underlying sexual dimorphism in infection responses (Duneau et al. BMC Biol 2017; Belmonte et al. Front Imm 2020)

Within-host pathogen dynamics

- Demonstrated that infection must be analyzed as sequential steps to understand outcomes (Duneau et al. eLife 2017, 2025; invited review Curr Opin Insect Sci 2022)
- Revealed that distinct infection stages impose evolutionary trade-offs on bacterial adaptation (Faucher et al. mBio 2021)
- Identified the first molecular mechanism underlying host-parasite coevolution (Duneau et al. BMC Biol 2011)

Genetic architecture of quantitative traits

- Characterized genetic basis of insecticide resistance (Duneau et al. G3 2018) and phenotypic plasticity in body size, gut morphology, and pigmentation (Lafuente et al. PLoS Gen 2018; Bonfini et al. eLife 2021; Lafuente et al. Mol Ecol 2024)

Evolutionary ecology

- Demonstrated that seminal fluid proteins function as a source of information to females and their effects are better understood as female adaptations than as male manipulation (Michalak et al. Am Nat 2026, BioRxiv)
- Showed cancer increases predation risk using *Drosophila* genetic tools (Duneau et al. Anim Behav 2022)
- Revealed parasites manipulating host behavior can escape host predators (Ponton et al. Nature 2006)

Academic leadership

- **Reviewer for 25 Scientific Journals (alphabetical order)**
  ◘ Animal Behaviour ◘ American Naturalist ◘ Applied Environmental Microbiology ◘ Biology Letters ◘ BMC Biology ◘ BMC ecology ◘ BMC Evolutionary Biology ◘ Coevolution ◘ Ecology and Evolution ◘ Evolution ◘ Epidemiology and Infection ◘ Heredity ◘ Immunity & Ageing ◘ Invertebrate Biology ◘ Invertebrate Survival Journal ◘ Journal of Animal Ecology ◘ Journal of Experimental Biology ◘ mSphere ◘ Nature Communications ◘ Oecologia ◘ Oikos ◘ Phil. Transactions of the Royal Society ◘ PLoS ONE ◘ PLoS Pathogens ◘ PNAS ◘ Proc Roy Soc B.
- **Reviewer for 5 funding agencies**
  Carnegie Trust for the Universities of Scotland ◘ Sigma Xi awards research grants program ◘ National Commission for Scientific and Technological Research of Chile ◘ National Fund for Scientific Research of Belgium (NFWO) ◘ European Research Council (ERC).
- **External expert for 1 PhD Viva and 3 thesis committees**
  - Thesis committee Lennard P at Univ. Edinburgh (UK) supervised by Prof. Dorin J (2022)
  - Thesis committee Rubinic M at Max Planck Institute for Infection Biology (Germany) supervised by Dr. Iatsenko I (2021, 2023, 2024)
  - Viva external examiner for Pinaud S at Univ. Perpignan (France) supervised by Dr. Gourbal B (2018)
  - Thesis committee Hanson M at EPFL (Switzerland) supervised by Prof. Lemaitre B (2018)
- **Administrative & organisational roles**
  - Part of the scientific committee of the French Network "Réseau Ecologie des Interactions Durables" (REID) (2020-2022).
  - Organisation of the National meeting of the French Network "Réseau Ecologie des Interactions Durables" (REID) (2017, Museum natural history of Toulouse).
  - Organisation of departmental seminars (2015 to 2017, University of Toulouse 3)
  - Organisation of seminars aiming to present experiments or analyses before they were performed. (From fall 2009 to spring 2011, Zoological Institute Basel)

Student supervision

- **PhD students (5)**
  - Main supervisor (Co-supervised with Dr. Beldade, Lisbon Univ.) of Yara Santos Rodrigues (2015 - 2020). She is now the Director of Science Promotion at Cabo Verde University and received the **Cabo Verde Prize for Young Scientists**.
  - Main supervisor (Co-supervised with Prof. Ferdy J-B, Toulouse Univ.) of Piotr Michalak (2021-2024)
  - Mentoring of 3 PhD students: Two from the Institute of Immunology and Infection Research in Edinburgh (Mary-Kate Corbally, 2019 –2022, Rebecca Belmonte, 2019 – 2023); and one at the University of Lisbon (Elvira Lafuente, 2013 – 2017).

- **Undergraduate and Master students (22)**
  I have supervised 15 undergraduate students and 7 master students since 2009. Eleven have been associated to publications, two of those students are first author of publications where I am last and corresponding author.

Teaching

I aim to share my passion for quantitative biology and experimental design with students, especially those less familiar with quantitative approaches (e.g. students from biomedical backgrounds). My biggest achievement in teaching is my ability to pass on my knowledge and passion of biological processes to my students via statistics and programming. I derive great satisfaction in knowing that I have helped many overcome their fear of using statistics. Some (non-exhaustive) examples below;

- "Evolution of the immune system" (2024- annual guest lecture) Level: Final year undergraduate, University of Edinburgh.
- "*Environment*" (2023), **P**roblem **B**ased **L**earning (PBL) & **H**ypothesis-**D**riven **D**evelopment sessions at Level: 1st year undergraduate, University of Edinburgh
- "*Field course in Ecology*" (2022) Level: 2nd year undergraduate, University of Edinburgh
- "*Evolutionary biology and genetics*" (2017 & 2018) Level: 3rd year undergraduate, University of Toulouse 3.
- Invited lecture on host-parasite coevolution (2017). Level: high school teachers.
- Summer school LabEx TULIP (2015) on *Integrative Ecology and Biology* "Biological interactions from genes to ecosystems". Level: from 4th year undergraduates to Postdoc.
- Departmental *workshop on statistical analysis of experimental data*. Level: Postgraduates (2014).

Scientific communications

- **Invited talks** (21)
  - Edinburgh fly community, University of Edinburgh, UK 03/2026 (talk, invited by Barry Denholm)
  - Centre for Ecology, Evolution, Environment changes, University of Lisbon, 12/2025 (talk)
  - Centre for Cardiovascular Sciences (CVS), University of Edinburgh, 04/2025 (talk, invited by Lizzie Haythorne)
  - Institute of Biology - Zoology, Freie Universität Berlin, 11/2020 (online talk; invited by Olivia Judson, Jens Rolf and Sophie Armitage)
  - Seminar New voices in Infection Biology, Max Planck Institute for Infection Biology, Berlin, 10/2020 (online talk; invited by Igor Iatsenko) **Recording at:** https://youtu.be/e0N7eg-U0hI
  - Department DGIMI, Montpellier University, 10/2020 (online talk; invited by Alain Givaudan)
  - Innsbruck University, Innsbruck, Austria- 11/2019 (talk; invited by Markus Möst)
  - Edinburgh University, Edinburgh, UK - 06/2018 (talk; invited by Sarah Reece)
  - EPFL, Lausanne, Switzerland -04/2018 (talk; invited by Bruno Lemaitre)
  - University of Burgundy, Dijon, FR - 12/2017(talk; invited by Thierry Rigaud)
  - University of Montpellier (SEEM), Montpellier, FR - 12/2017(talk; invited by Karen McCoy)
  - Insect Biology Research Institute, Tours, FR - 10/2017 (talk; invited by Joel Meunier)
  - CNRS, Gif-sur-Yvette, FR - 04/2016 (talk; invited by Frédéric Mery)
  - Centre Biologie du Développement, Toulouse, FR - 04/2016 (talk; invited by Alain Vincent)
  - Conference LabEx TULIP, Toulouse, FR - 03/2016 (talk; invited by Etienne Danchin)
  - Institute for advanced study, Toulouse, FR - 06/2015 (talk; invited by Arnaud Togneti)
  - Center for infectious disease dynamics, PennState University, University Park, PA, USA - 04/2014 (talk; invited by David Hughes)

  - Department of Evolution, Ecology and Genetics, Australian National University, Canberra, Australia - 02/2014 (talk; invited by Hanna Kokko).
  - Department of Ecology and Evolutionary Biology, Rochester University, Rochester, USA - 11/2013 (talk; invited by John Jaenike).
  - Department of Evolutionary Biology of Cornell University, Ithaca, USA - 2012 (talk)
  - *Institute for Development Research* (IRD), Montpellier, FR - 2010 (talk; invited by Karen McCoy)
- **Conferences** (19)
  - Ecological immunology 2025, Blossin, Germany (Poster)
  - SIGNET 2025, Perth, UK - 04/2025 (Selected oral presentation)
  - Scottish Drosophila meeting – 03/2025 (Selected oral presentation)
  - DUKPC 2025, Glasgow, UK - 02/2025 (Poster)
  - Conference ESEB (2nd joint congress), Montpellier, FR - 08/2018 (Poster)
  - Conference Jacques Monod "Open questions in ecology and evolution in infectious diseases: from fundamental research to evolutionary medicine" - Roscoff Biological Station, FR - 10/2017 (Poster)
  - Conference Immuninv2017, Lyon, FR - 06/2017 (Contributed talk)
  - REID Annual Conference, Poitiers, FR - 03/2016 (Contributed talk)
  - Conference 15th ESEB, Lausanne, Switzerland - 08/2015 (Contributed talk)
  - Conference Jacques Monod, "Infectious diseases as drivers of evolution: the challenges ahead" - Roscoff Biological Station, FR - 09/2014 (contributed talk)
  - Drosophila research conference, San Diego, USA - 03/2014 (poster)
  - Conference 14th ESEB, Lisbon, Portugal - 08/2013 (poster)
  - Drosophila research conference, Washington DC, USA - 03/2013 (poster)
  - Conference ESEB (joint congress), Ottawa, Canada - 08/2012 (contributed talk)
  - Conference 13th ESEB, Tubingen, Germany - 08/2011 (contributed talk)
  - Conference Swiss-Russian Seminar, Freiburg, Switzerland - 2010 (contributed talk)
  - Conference 16th EMPSEB, Wierzba, Poland - 2010 (contributed talk)
  - Conference 12 ESEB, Turin, Italy - 2009 (contributed talk)
  - Conference 15th EMPSEB, Shoorl, Netherlands - 2009 (contributed talk)

Summary of publications

Total: 38 peer-reviewed publications of which 19 first/last/corresponding
+ 5 in BioRxiv (5 first/last/corresponding)
+ 4 yet unpublished thesis chapters (4 first/last/corresponding)

Google scholar profile: https://scholar.google.fr/citations?user=VhsB4z0AAAAJ&hl=en

- H-index: 22
- Total citations: ~1950 (31 publications with more than 10 citations)

Complete publication record

Below I have listed my publications from the most recent to the oldest in the following categories:

- Publications with a leading role
- Publications with a supporting role (where I indicate my contribution)
- Publications in bioRxiv
- Thesis chapters or exist as draft (as they reflect future publications)

Publications with a leading role († corresponding author; * equal contribution)

2021 to date:

1. Michalak P, J-B Ferdy†*, Duneau D†* *The role of fecundity dynamics and time of injury for female reproductive fitness.* **Roy. Soc. Open Science**

We studied experimentally the impact of wounds on reproduction and tested the hypothesis that injury at different periods of the fecundity dynamics will have different impact on fitness. We found that females were able to quickly compensate for the short-term consequences of wounding.

2. Michalak P, **Duneau D†*,** Ferdy J-B†*. *Seminal fluid proteins as regulation factors for optimizing reproduction: a modeling approach*. **American Naturalist** doi.org/10.1086/740809

Developed a biologically informed theoretical model showing how seminal fluid proteins synchronize sperm and egg release, reducing unfertilized egg production. Identified SP exhaustion as the key synchronization signal and demonstrated that sexual conflict over SP-mediated regulation of female physiology should be limited, though conflict over optimal remating rate remains possible.

3. Jerónimo M, Garizo A, Atencio G, **Duneau D†***, Beldade P†*. (2025) *Wound-induced eyespots on butterfly wings at the intersection of immune response and pigmentation development* **BMC Biology** doi.org/10.1186/s12915-025-02397-3

Showed that immune activation modulates wound-induced ectopic eyespot formation in butterfly wings, revealing crosstalk between immune function and pigmentation development.

4. **Duneau D**†*, Lafont P*, Lauzeral P, Parthuisot N, Faucher C, Xuerong J, Buchon N, Ferdy JB†. (2025) *A within-host infection model to explore tolerance and resistance.* **eLife** doi.org/10.7554/eLife.104052

Developed a within-host infection model integrating theory and experiments to distinguish between tolerance and resistance mechanisms. Provided a practical experimental framework for explaining individual differences in susceptibility to infection, a methodological foundation applied in subsequent work (e.g. Monteiro-Black et al. 2025).

5. Lafuente E, **Duneau D**, Beldade P. (2024) *Genetic architecture of plasticity for pigmentation components in* Drosophila melanogaster. **Molecular Ecology** doi.org/10.1111/mec.17294

Applied genome-wide association mapping in Drosophila melanogaster to dissect the genetic basis of temperature-dependent pigmentation plasticity. Found that different body parts rely on largely independent genetic architectures, challenging the assumption of a common genetic basis for pigmentation plasticity. Contribution: I supervised the student for the genomic analysis and for the validation of allele candidate with functional genetics.

6. **Duneau† D**, Möst M, Ebert D. (2022) *Evolution of sperm morphology in* Daphnia *species.* **Peer Community Journal** (Diamond Scientific Publishing) doi.org/10.24072/pcjournal.182

Based on a recent phylogeny, we identified that *Daphnia* had among the smallest recorded sperm and studied the evolution of sperm length in this clade.

7. **Duneau† D**, Buchon N (2022) *Gut cancer increases the risk for* Drosophila *to be preyed upon by hunting spiders*. Accepted in **Animal Behaviour** doi.org/10.1016/j.anbehav.2022.07.003

Diseased individuals are more preyed upon than healthy individuals. Although this is intuitive, this conventional wisdom has been subjected more to speculation than to empirical study. To test the idea, we genetically induced colon cancer in *Drosophila* and found that cancerous flies are more frequently predated than healthy ones.

8. **Duneau† D**, Ferdy J-B (2022) *Pathogen within-host dynamics and disease outcome: what can we learn from insect studies?* Accepted in **Current Opinion in Insect Science** doi.org/10.1016/j.cois.2022.100925

I have been invited by Sophie Armitage (University of Berlin) and Barbara Milutinović (University of Münster) to review the latest knowledge on the connection between within-host pathogen dynamics and disease outcome.

9. Rodrigues YK, van Bergen E, Alves F, **Duneau D***, Beldade P* (2021) *Additive and non-additive effects of day and night temperatures on thermally plastic traits in a model for adaptive seasonal plasticity.* **Evolution** evo.14271

We tested the effects of circadian temperature fluctuations on a series of thermal plasticity traits in a model of adaptive seasonal plasticity, the *Bicyclus anynana* butterfly.

10. Faucher C*, Mazana V*, Kardacz M, Parthuisot N, Ferdy J-B, **Duneau† D** (2021) *Step-specific adaptation and trade-off over the course of an infection by GASP-mutation small colony variants.* **mBio** doi.org/10.1128/mBio.01399-20

Within-host bacterial adaptations are generally focused on antibiotic resistance, rarely on the adaptation to the environment given by the host, and the potential trade-off hindering adaptations to each step of the infection are rarely considered. Using *Drosophila melanogaster* as host and the bacteria *Xenorhabdus nematophila*, we studied those trade-offs that are key to understand intra-host evolution, and thus the dynamics of the infection. We found that adaptation to late infection trade-off with adaptation to early infection.

2020 and before:

11. Belmonte RL, Corbally M-K, **Duneau D**[*†], Regan JC[*†] (2020) *Sexual dimorphisms in innate immunity and responses to infection in* Drosophila melanogaster. **Frontiers in Immunology** doi.org/10.3389/fimmu.2019.03075

We evaluated the sexual dimorphism of *Drosophila* immune responses based on an exhaustive survey of studies on the immunity of this model. We highlighted that current theories are too generalist to have sufficient explanatory power, and so, generated a set of realistic, testable, hypotheses from which my project derives.

12. **Duneau D**[†*], Sun H[*], Revah J, San Miguel K, Kunerth HD, Caldas IV, Messer PW, Scott JG, Buchon N (2018) *Genome wide analysis of resistance to an organophosphate and a pyrethroid insecticide.* **G3: Genes|Genomes|Genetics** doi.org/10.1534/g3.118.200537

Using GWAS with the Drosophila Reference Genetic Panel (DGRP) found the genetic basis of the resistance to Parathion and Deltamethrin, two commonly used insecticides. We showed that commonly used insecticides have a strong impact on the evolution of non-target species such as *Drosophila* and that resistance alleles are present worldwide.

13. Lafuente E, **Duneau D**, Beldade P (2018) *Genetic basis of thermal plasticity variation in* Drosophila melanogaster *body size*. **PLoS Genetics** doi.org/10.1371/journal.pgen.1007686

Using GWAS in *Drosophila*, we determined the genetic basis of thermal plasticity of thorax and abdomen size. Contribution: I supervised the experimental design, the genomic analysis and the validation of allele candidates with functional genetics.

14. **Duneau**[†] **D**, Lazzaro B (2018) *Persistence of an extracellular systemic infection across metamorphosis in a holometabolous insect.* **Biology Letters** doi.org/10.1098/rsbl.2017. 0771

We showed that systemic infection with an extracellular bacterium can transverse life stages.

15. **Duneau**[†] **D**, Ferdy JB, Revah J, Kondolf HC, Ortiz GA, Lazzaro BP, Buchon N (2017) *Stochastic variation in the initial phase of bacterial infection predicts the probability of survival in* D. melanogaster. **eLife** doi.org/10.7554/eLife.28298 (F1000Prime RECOMMENDED)

A central problem in biomedicine is understanding why individuals exposed to seemingly identical infections can have radically different outcomes. Using Drosophila melanogaster, we combined functional genetics and mathematical modeling to analyze the main determinants of stochastic infection outcome. We showed that even without genetic variance and in a controlled environment, some hosts survive while others succumb. We characterised two new bacterial load measures (load upon death and load during chronicity) to describe the infectious process, and demonstrated that host survival is predicted by within-host bacterial proliferation, with small variations in innate immune response differentiating survivors from non-survivors. We developed an integrative dynamic model that accurately predicts infection outcome and shows that the timing of immune response initiation determines survival. My research program is largely built on the knowledge and methods acquired during this study. ("Highlight" by Graham and Tate eLife 2017 6:e32783; recommended 4× "very good" on Faculty Opinions, https://archive.connect.h1.co/article/731979509)

16. **Duneau**[†] **D**, Kondolf HC, Im JH, Ortiz GA, Chow C, Fox MA, Eugénio AT, Buchon N, Lazzaro BP (2017) *The Toll pathway underlies host sexual dimorphism in resistance to both Gram-negative and positive-bacteria in* Drosophila. **BMC Biology** doi.org/10.1186/s12915-017-0466-3

We elucidated part of the mechanisms underlying the difference between sexes in infection outcome in *Drosophila.* We revealed that the Toll pathway, responsible for the sexual dimorphism in some human infectious diseases (e.g. HIV), also explains why female *Drosophila* are slower than males to control pathogen proliferation.

17. **Duneau**[†] **D**, Ebert D, Du Pasquier L (2016) *Infections by* Pasteuria *do not protect its natural host* Daphnia magna *from subsequent infections.* **Developmental & Comparative Immunology** doi.org/10.1016/ j.dci.2015.12.004

We tested and concluded that *Daphnia* do not have immunological memory upon bacterial infections.

18. **Duneau**[†] **D**, Ebert D (2012) *Host sexual dimorphism and parasite adaptation.* **PLoS Biology** doi.org/10.1371/journal.pbio.1001271

In this "essay" we propose for the first time the idea that the sexual dimorphism of diseases may be the result of the specific adaptation of parasites to the sex of their host. Similarly, as organisms adapt to the environment to which they are most frequently exposed, parasites can adapt to the sex they encounter most frequently (e.g., either because males and females are exposed differently, or because one sex is more easily infected than another due to immune differences). As a result, parasites behave differently depending on the sex they infect.

19. **Duneau**† **D**, Luijckx P, Ruder L, Ebert D (2012) *Sex-specific effects of a parasite evolving in a female-biased host population.* **BMC Biology** doi.org/10.1186/1741-7007-10-104

We have shown here that a parasite can be better adapted to the sex it encounters most frequently, empirically supporting the hypothesis proposed Duneau & Ebert PLoS Biology 2012.

20. **Duneau**† **D**, Ebert D (2012) *The role of molting in parasite defense.* **Proceedings of the Royal Society of London B** doi.org/10.1098/rspb.2012.0407

We show that moulting is not only a weakness but can be beneficial to prevent infection by shedding bacteria. (Covered by: Le Figaro (https://bit.ly/2tbIPUk) & Live Science (https://bit.ly/37aT6Po)

21. **Duneau**† **D**, Luijckx P, Ben-Ami F, Laforsch C, Ebert D (2011) *Resolving the infection process reveals striking differences in the contribution of environment, genetics and phylogeny to host-parasite interactions.* **BMC Biology** doi.org/10.1186/1741-7007-9-11

We investigated the mechanism of infection underlying coevolution between a host (*Daphnia magna*) and his parasite (*Pasteuria ramosa*). We found that the specificity depends on the capacity of the parasite to attach or not to the host oesophagus. I published here the "attachment test" method which is still used routinely in my PhD supervisor lab to quickly determine the ability of the bacteria of a given genotype to infect a given host genotype. It is the first study to describe a mechanism responsible for the coevolution between an animal host and its parasite.

22. Ponton* F, **Duneau* D**, Sanchez M, Courtiol A, Terekhin A, Budilova EV, Renaud F, Thomas F (2009) *Effect of parasite-induced behavioral alterations on juvenile development.* **Behavioral Ecology** doi.org/10.1093/beheco/arp092

We studied whether the juveniles of manipulated female *Gammarus* (i.e. with a high risk of predation) developed more rapidly in their mother's ventral pouch, a behaviour that could have been selected to reduce the juveniles' chances of predation; this is not the case.

23. **Duneau D**, Boulinier T, Gomez-Diaz E, Petersen A, Tveraa T, Barrett RT, McCoy KD (2008) *Prevalence and diversity of Lyme borreliosis bacteria in marine birds.* **Infection, Genetics and Evolution** doi.org/10.1016/j.meegid.2008.02.006

We studied the diversity of *Borrelia spp*. circulating in seabirds. Our findings indicate that seabirds may be an important component in the global epidemiology and evolution of Lyme disease.

Publications with a supporting role

2021 to date:

24. Bonfini A, Dobson AJ, **Duneau D**, Revah J, Liu X, Houtz P, Buchon N (2021) *Diet composition plastically resizes the* Drosophila *midgut by affecting cell gain and loss, stem cell-niche coupling and enterocyte size.* **eLife** doi.org/10.7554/eLife.64125

We studied the phenotypic plasticity of *Drosophila* gut in response to glucose level in diet. Contribution: Performed the GWAS and all the statistical analyses of the paper.

2020 and before:

25. Bento G, Fields P, **Duneau D**, Ebert D (2020) *An alternative route of bacterial infection is associated with a polymorphism at an alternative resistance locus.* **Heredity** doi.org/10.1038/s41437020 -0332-x

Contribution: I discovered the alternative route of infection.

26. Pineaux M, Merkling T, Danchin E, Hatch S, **Duneau D**, Blanchard P, Leclaire S (2020) *Sex and hatching order modulate the association between MHC-diversity and fitness in early-life stages of a wild seabird.* **Molecular Ecology** doi.org/10.1111/mec.15551

Contribution: Helped in the analyses and writing as an evolutionary parasitologist knowing the immune system and the sexual dimorphism in diseases.

27. Corse E, Tougard C, Archambaud G, Agnèse J-F, Messu Mandeng FD, Bilong Bilong CF, **Duneau D**, Zinger L, Chappaz R, Xu CCY, Méglecz E, Dubut V (2019) *One-locus-several-primers: a strategy to improve the taxonomic and haplotypic coverage in diet metabarcoding studies.* **Ecology & Evolution** doi.org/10.1002/ece3.5063

Contribution: Sampled spider webs in the tropical rainforest of French Guyana to show that we used them as DNA traps to describe biodiversity with metabarcoding.

28. Ebert D, **Duneau D**, Hall M, Luijckx P, Andras J, Du Pasquier L, Ben-Ami F (2016) *A population biology perspective on the stepwise infection process of the bacterial pathogen* Pasteuria ramosa *in* Daphnia. **Advances in parasitology** doi.org/10.1016/bs.apar.2015.10.001

We demonstrate that a population biology approach taking into consideration the natural genetic and environmental variation at each step of the infection can greatly aid our understanding of the evolutionary processes shaping disease traits. Contribution: This manuscript is largely based on my PhD thesis.

29. Avila F, Cohen A, Ameerudeen F, **Duneau D**, Suresh S, Mattei A, Wolfner M (2015) *The D*rosophila *mating plug protein, PEBme, is required to maintain the ejaculate within the female reproductive tract at the termination of copulation.* **Genetics** doi.org/10.1534/genetics.115. 176669

Contribution: Showed with macrophotography that the inability of females to subsequently retain the ejaculate in their reproductive tracts after mating was frequent.

30. Luijckx P, **Duneau D**, Andras J, Ebert D (2014) *Cross-species infection trials reveal cryptic parasite varieties and a putative polymorphism shared among host species.* **Evolution** doi.org/10.1111/evo.12289

Contribution: Designed and did most experiments together with the first author.

31. Luijckx P, Fienberg H, **Duneau D**, Ebert D (2013) *A matching-allele model explains host resistance to parasites.* **Current Biology** doi.org/10.1016/j.cub.2013.04.064 (F1000Prime RECOMMENDED)

First times the genetic model of coevolution, the matching-allele model, was shown. Contribution: Performed all the attachment tests of this study (based on Duneau et al. BMC Biol. 2011).

32. Luijckx P, Fienberg H, **Duneau D**, Ebert D (2011) *Resistance to a bacterial parasite in the crustacean* Daphnia magna *shows Mendelian segregation with dominance.* **Heredity** doi.org/10.1038/hdy.2011.122

Contribution: Performed all the attachment tests of this study (based on Duneau et al. BMC Biol. 2011).

33. Ponton F, Otalora-Luna F, Lefevre T, Guerin PM, Lebarbenchon C, **Duneau D**, Biron DG, Thomas F (2011) *Water-seeking behavior in worm-infected crickets and reversibility of parasitic manipulation.* **Behavioral Ecology** doi.org/10.1093/beheco/arq215

Contribution: Participated in sampling and performed several experiments of the study.

34. Gómez-Díaz E, Doherty P Jr, **Duneau D**, McCoy KD (2010) *Cryptic vector divergence masks vector-specific patterns of infection: an example from the marine cycle of Lyme borreliosis.* **Evolutionary Applications** doi.org/10.1111/j.1752-4571.2010.00127.x

Contribution: Performed everything based on Sanger PCR.

35. McCoy KD, **Duneau D**, Boulinier T (2008) *Spécialisation de la tique des oiseaux marins et diversité des bactéries du complexe* Borrelia burgdorferi sensu lato*, agents de la maladie de Lyme : effets en cascade dans les systèmes à vecteur.* **Les actes du BRG** 277-291 (french publication with reviewing committee)

Summary of the work in Duneau et al IGE 2008 for the colloquium BRG.

36. Ponton F, Lebarbenchon C, Lefèvre T, Biron DG, **Duneau D**, Hughes DP, Thomas F (2006) *How parasitic Gordian worms cut the Gordian knot: a novel solution to predation upon the host.* **Nature** doi.org/10.1038/440756a

As prisoners in their living habitat, parasites should be vulnerable to the predators of their hosts. We show that the parasitic hairworm can escape from the predators of its hosts. Contribution: Performed all the experiments with frogs (one of the three predators).

37. Ponton F, Lebarbenchon C, Lefèvre T, Thomas F, **Duneau D**, Marché L, Renault L, Hughes DP, Biron DG (2006) *Hairworm anti-predator strategy: a study of causes and consequences.* **Parasitology** doi.org/10.1017/S0031182006000904

Contribution: I designed the whole nursery for the hairworms and monitored all the mating to measure the fitness.

38. Ponton F, Biron DG, Joly C, **Duneau D**, Thomas F (2005) *Ecology of populations parasitically modified: a case study from a gammarid (*Gammarus insensibilis*)-trematode (*Microphallus papillorobustus*) system.* **Marine Ecology-Progress Series** doi.org/ 10.3354/meps299205

We studied the consequence of behavioural manipulation by parasites on the ecology of host populations. Contribution: I was involved in most of the experiments and quantified all host energy resources (glucose, lipid, etc.).

Publications in bioRxiv († corresponding author; * equal contribution)

39. Michalak P, Duneau D†*, J-B Ferdy†* *Seminal fluid proteins can mitigate sexual conflict: the case of remating in insects.* doi.org/10.64898/2026.02.27.708467 (Submitted to **PNAS**)

We used a mathematical model of reproductive physiology, informed by accumulated knowledge on Drosophila melanogaster, to examine the role of seminal fluid proteins in sexual conflict over optimal remating rate. Our results suggest that while a conflict over the remating rate does exist, seminal fluid proteins reduce its intensity, highlighting their role in aligning the interests of both sexes.

40. Rubinic M, Arias-Rojas A, Martinez K, Klimek W, Paczia N, Alagesan K, Duneau D, Iatsenko I *Sex differences in Drosophila intestinal metabolism contribute to sexually dimorphic infection outcome and alter gut pathogen virulence.* doi.org/10.1101/2025.05.22.655590 (Second revision for **PNAS**)

We studied the sexual dimorphism of intestinal infections of D. melanogaster. We showed that males exhibit increased a key antioxidant defence system. It allows them to withstand oxidative stress-induced defecation blockage and clear the pathogen from the intestine, resulting in survival. We also showed that the bacteria showed increased expression of several virulence factors in female gut, indicating a change in pathogen behaviour depending on the host they infect (a theory I have been developing since Duneau & Ebert PLoS Biol. 2012). I have participated in developing the hypothesis, trained the first author to perform the Bulk RNAseq analysis and in using R. I also analysed the GWAS.

41. Monteiro-Black B, Corbally MK, Aleksandrowicz J, Taggart AM, Langella F, Belmonte B, Mika P, **Duneau D**†*, Regan J†* *Different ways to die: harnessing variation in Drosophila reveals loss of tolerance and resistance over age and sex-specific association between infection susceptibility and lifespan.* doi.org/10.1101/2025.07.07.663438

Using methods developed in Duneau et al. eLife 2017 and Duneau et al. eLife 2025, we showed that immune ageing can underlie a decrease in the ability to sustain pathogens, not only to control their proliferation.

42. **Duneau**†* **D**, Gallet R*, Adhiambo M, Delétré E, Chailleux A, Khamis F, Subramanian S, Brévault T, Fellous S† *Sex-specific sub-lethal effects of low virulence entomopathogenic fungi may boost the Sterile Insect Technique*. doi.org/10.1101/2024.03.07.583916

Our experimental study supports a paradigm shift in pest control from the "maximally virulent" approach to a "sex-dependent" strategy aimed at selecting bioagents for boosted Sterile Insect Techniques. Such an approach can maintain the competitiveness of sterile males to access the females, while contributing to horizontal transmission of pathogens which increases female mortality and decrease their fertility.

43. **Duneau**† **D**, Altermatt F, Ferdy J-B, Ben-Ami F, Ebert D. *Estimation of the propensity for sexual selection in a cyclical parthenogen*. doi.org/10.1101/2020.02.05.935148

Our data from the field show that sexual selection is present in *Daphnia*, although it is mainly parthenogenetic, and that this selection probably manifests itself through a combination of female choice and male competition.

Thesis chapters or exist as draft

1. Kambou S, Liu C, Tam-Hui T, Gineste A, **Duneau D**†*, Chassagne F†* *Using the Drosophila/Serratia marcescens system to test in vivo new antimicrobial compounds.* (First revision for **Applied and Environmental Microbiology**)

We used my expertise in *Drosophila* infections and the expertise of François Chassagne in ethnopharmacology to successfully develop a method of *in vivo* screening of antimicrobial compounds present in traditional medicine.

2. Rodrigues YK, **Duneau D**†*, Beldade P†*. *Seasonal and sexual dimorphism in immunity in a thermal plasticity model.*

Using methods developed in Duneau et al. eLife 2017 with *Drosophila*, we studied the thermal phenotypic plasticity of the immune system of the butterfly *Bicyclus anynana*.

3. Corbally MK, **Duneau D***, Regan J* *Investigating the role of a putatively longevity-associated transposable element in lifespan and immunity*.
   Using GWAS with transposable element (TE), instead of SNP, in *Drosophila*, we discovered that a TE inclusion in regulatory region of major immune gene affected lifespan and the response to infection.

4. Xiao-Li Bing*, Peter Nagy*, **Duneau D***, Alessandro Bonfini, Katia Troha, Nicolas Buchon. *Deciphering similarities and differences between Drosophila melanogaster and Drosophila suzukii in response to bacterial infections*.

We studied the response of infection between a fruit pest *Drosophila* (*D. suzukii*) and the reference model organism *D. melanogaster*. We showed a surprising similarity, suggesting that a large part of the knowledge on *D. melanogaster* could be used to fight a very important crop pest. I have been involved in the analysis of the bacterial within-host dynamics and in the interspecies comparison of the transcriptomic response, which brings overlooked technical challenges.